What changed in ansible-core 2.19 and 2.20 for this course

- **Course:** Ansible for DevOps automation mastery (Arabic), DevCloudLab
- **Applies to:** every lecture. Attach to "The hosts file" (Section 2).
- **Updated:** November 2026, checked against ansible-core 2.20.

> **ملخص سريع بالعربي:** الكورس اتسجل على إصدار أقدم من ansible-core. الفيديوهات لسه صحيحة في الفكرة، بس فيه شوية تفاصيل اتغيرت في 2.19 و 2.20:
>
> 1. في ملف الـ hosts استخدم `ansible_ssh_private_key_file` (مش `ansible_private_key`).
> 2. اكتب الـ facts بالشكل ده: `ansible_facts['distribution']` بدل `ansible_distribution`.
> 3. جهاز التحكم (control node) محتاج Python 3.12 أو أحدث، والسيرفرات محتاجة Python 3.9 أو أحدث.
> 4. الشروط في `when` لازم ترجع true أو false بشكل صريح.
>
> كل ملفات الكورس بعد التعديل موجودة في مستودع الـ lab.

1. Check your version

```
ansible --version        # look at "ansible [core 2.xx]" and "python version"
python3 --version        # control node: 3.12 or newer for ansible-core 2.20
```

If you installed with `pip install ansible` in 2026 you have ansible-core 2.20. Ubuntu 24.04's `apt install ansible` gives an older 2.16, where the videos work exactly as recorded. A virtual environment is the cleanest way to get 2.20:

```
sudo apt install -y python3-venv
python3 -m venv ~/ansible-env && source ~/ansible-env/bin/activate
pip install ansible
```

2. Python versions

	ansible-core 2.19	ansible-core 2.20
Control node (where you run `ansible-playbook`)	Python 3.11+	**Python 3.12+**
Managed hosts (the Multipass VMs)	Python 3.8+	**Python 3.9+**

Ubuntu 24.04 ships Python 3.12, so the course VMs are fine both as control node and as targets. On macOS, use Homebrew Python 3.12+ or `pipx install ansible`.

3. Inventory: the SSH key variable (lecture "The hosts file")

This is the change that affects students most.

Variable	What it means	Use it?
`ansible_ssh_private_key_file`	**Path** to the private key file	**Yes, recommended**
`ansible_private_key_file`	Same thing, generic name	Yes
`ansible_private_key` / `ansible_ssh_private_key`	**New in 2.19:** the key **contents** (the text of the key), loaded into ssh-agent	No, not for this course

If you write a **path** into `ansible_private_key`, ansible-core 2.19+ treats it as key contents and fails with:

```
Failed to authenticate: Failed to add configured private key into ssh-agent: Cannot
utilize private_key with SSH_AGENT disabled
```

Fix: rename the variable to `ansible_ssh_private_key_file`.

```
[my_hosts:vars]
ansible_connection=ssh
ansible_user=ansibleadmin
ansible_ssh_private_key_file=~/.ssh/id_rsa
```

Two related clean-ups used in the lab repo:

- The connection variable is `ansible_connection=ssh` (not `ansible_connect`). Ansible silently ignores unknown variables, so a typo here doesn't raise an error, but it does nothing.
- `host_key_checking` is an `ansible.cfg` setting (`[defaults] host_key_checking = False`). It isn't an inventory variable.

4. Facts: use `ansible_facts['...']`

The videos use injected fact variables such as `ansible_distribution`, `ansible_nodename` and `ansible_distribution_release`. In 2.20 this style is **deprecated** (setting `INJECT_FACTS_AS_VARS`), and its default is scheduled to switch off in ansible-core 2.24. It still works today but prints a deprecation warning. The future-proof form is:

Video	Write this instead
`{{ ansible_nodename }}`	`{{ ansible_facts['nodename'] }}`
`{{ ansible_distribution }}`	`{{ ansible_facts['distribution'] }}`
`{{ ansible_distribution_version }}`	`{{ ansible_facts['distribution_version'] }}`

Video	Write this instead
`{{ ansible_distribution_release }}`	`{{ ansible_facts['distribution_release'] }}`
`{{ ansible_os_family }}`	`{{ ansible_facts['os_family'] }}`

Connection variables such as `ansible_host`, `ansible_user` and `inventory_hostname` aren't facts and don't change.

5. Templating and conditionals (2.19 overhaul)

ansible-core 2.19 rewrote the templating engine. The changes you may notice in the labs:

- **Conditionals must be real booleans.** If `deploy_custom_page: "false"` is a *string* (quoted, or passed with `-e deploy_custom_page=false`), `when: deploy_custom_page` now fails with: `Conditional result (True) was derived from value of type 'str' ... Conditionals must have a boolean result`. The "(True)" is the trap: a non-empty string is truthy, so older versions silently ran the task. Write `when: deploy_custom_page | bool`, or use a real YAML `true`/`false`. (A temporary escape hatch is `ALLOW_BROKEN_CONDITIONALS=True`.)
- **Template syntax errors are caught.** An unclosed comment such as `{# templates/App.js.j2}` (no closing `#}`) now fails with "Missing end of comment tag". The lab repo's `App.js.j2` is fixed.
- **No more "double templating".** A variable whose value contains `{{ ... }}` from an untrusted source isn't rendered a second time. None of the course labs depend on this.
- **Better error messages** show the file, line and column. Read them, because they usually point straight at the problem.

6. Other changes you may hit

Area	Change	What to do
Module names	Short names (`apt`, `copy`) still work	The lab repo uses FQCNs (`ansible.builtin.apt`, `community.mongodb.mongodb_user`) so ansible-lint is happy
Collections	`pip install ansible-core` alone has no `community.*` collections	`ansible-galaxy collection install -r requirements.yml` in each lab folder
Galaxy	The v2 Galaxy API was removed in 2.20. The instructor's demo role was deleted from Galaxy after the lecture	Publish your own role, or install from GitHub (see `section03-roles/apache-playbook/requirements.yml`)
`include_vars`	`extensions` and `ignore_files` must be lists	Not used in the labs
Transport	`DEFAULT_TRANSPORT=smart` was removed	Use `ssh` (the default)

Area	Change	What to do
Windows	PowerShell no longer strips quotes from module arguments	Quote `win_shell` and `win_command` arguments the way PowerShell expects
`become`	A become timeout now reports the host as **unreachable** instead of failed	Nothing to change
`bool` filter	Passing values that aren't real booleans or yes/no strings is deprecated	Keep values to `true`/`false`/`yes`/`no`

7. Changes outside Ansible that affect the MERN and LAMP projects

- **create-react-app** was deprecated by the React team in 2025. `npx create-react-app` still works but prints warnings. The lab keeps it so it matches the video. For new projects, use Vite (`npm create vite@latest`).
- **MongoDB 8.0** on Ubuntu 24.04: the repo uses `ansible.builtin.deb822_repository` and the `mongosh` shell (`~/.mongoshrc.js`, not the old `~/.mongorc.js`).
- **MySQL 8.4** (LAMP assignment): the `mysql_native_password` plugin is disabled by default. Use the default `caching_sha2_password` with a current PyMySQL.

8. Where are the corrected files?

The lab repo has every playbook, role, template and web asset from Sections 1 to 6, updated for ansible-core 2.20. It also includes `App.css` for "Creating the web server 01". The `README.md` maps each folder to its lecture and lists every fix made compared with the videos.